



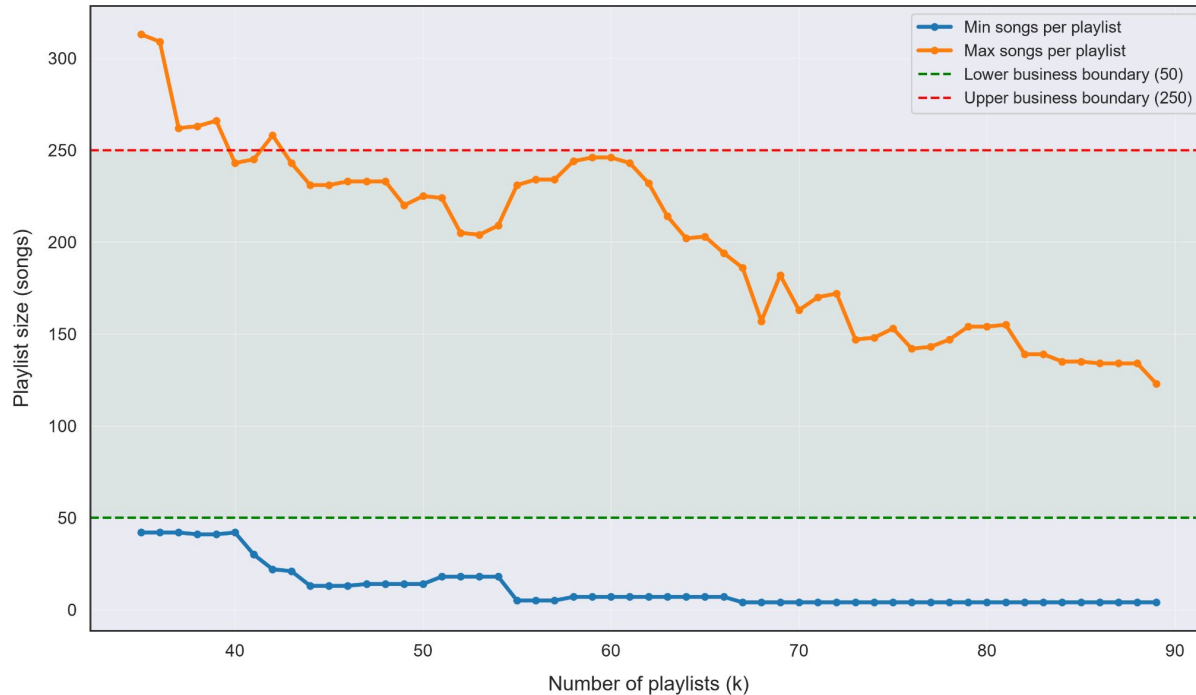
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Spotify Playlist Creation - Mood Based

Overall Takeaway



Playlists can be created for the given songs and using the relevant features, however, mathematical computations do not meet the defined business boundaries/ needs



Prototype - Inside the Build



1. Data Input

Spotify Songs

2. Machine Learning

Cluster few playlists based on all relevant features

- 8 playlists

Select 3 musical mood features (research and variance based)

- energy
- valence
- acousticness

3. Operator Input

Define & apply business boundaries

- median = 200 songs
- k = 27 playlists

4. User/Expert Input

Manually verify playlist assignment

Playlist Verification



Exemplary sample playlists set similar moods → GOAL ACHIEVED

Playlist 1 - high mood score

- O Meu Amor É Glòria - Sílvia Pérez Cruz
- Previsao - Bossacucanova
- Aqua De Coco - Marcos Valle
- Samba de Verão - Marcos Valle
- Aquarela Do Brasil - Gal Costa

Playlist 2 - low mood score

- Different Pulses - Asaf Avidan
- Devil Or Angel - Lou Doillon
- Inbetween Days - The Cure
- L'ós - La iaia
- I Didn't See It Coming (Richard X Mix) - Belle & Sebastian

Way Forward



The applied methods do not in their entirety and single-handedly provide answers to the current business needs.

→ Introduce more features to the clustering process (a music expert opinion is required here)

- Mood playlists should be separated from style/genre based playlists
- Songs of the same style/genre do not necessarily have the same mood

→ Test the current playlists and allow customisation by the users

The results from above should inform a strategic way forward



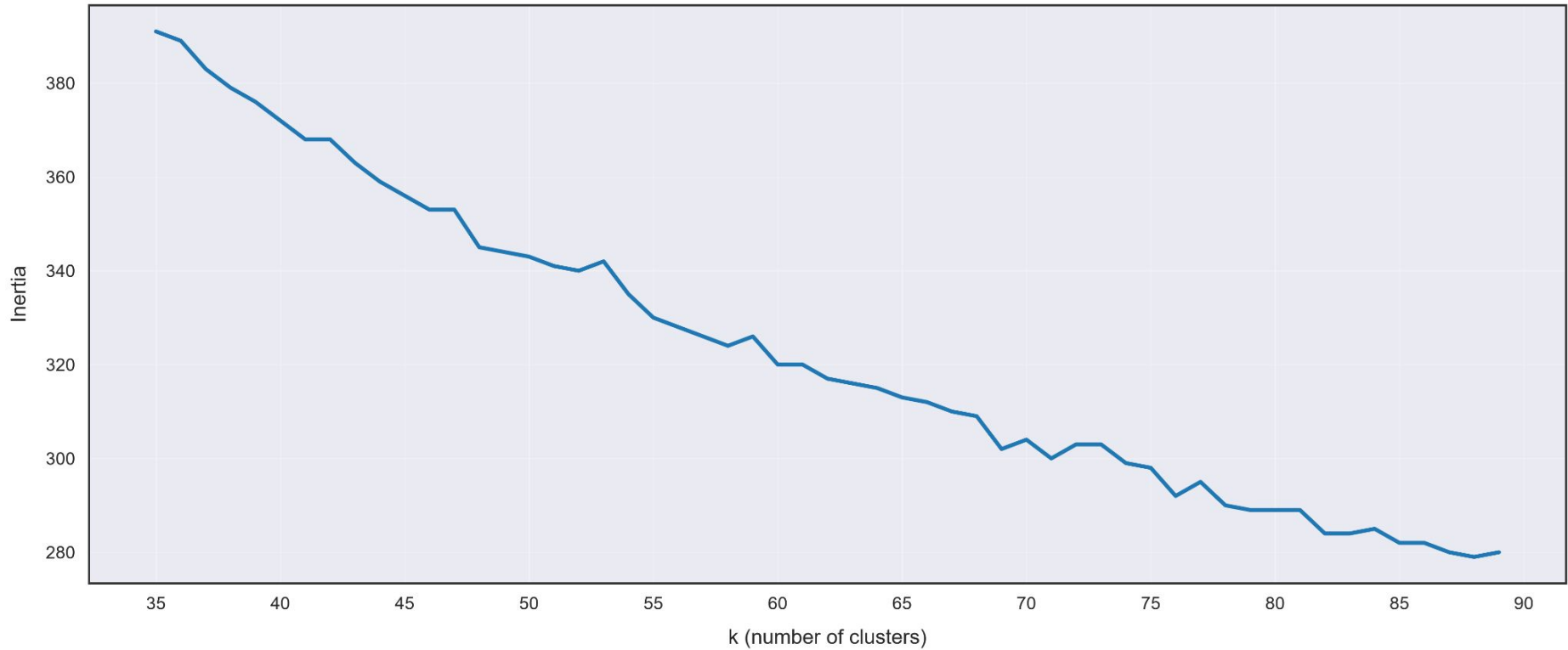
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Extra Slides

Elbow method



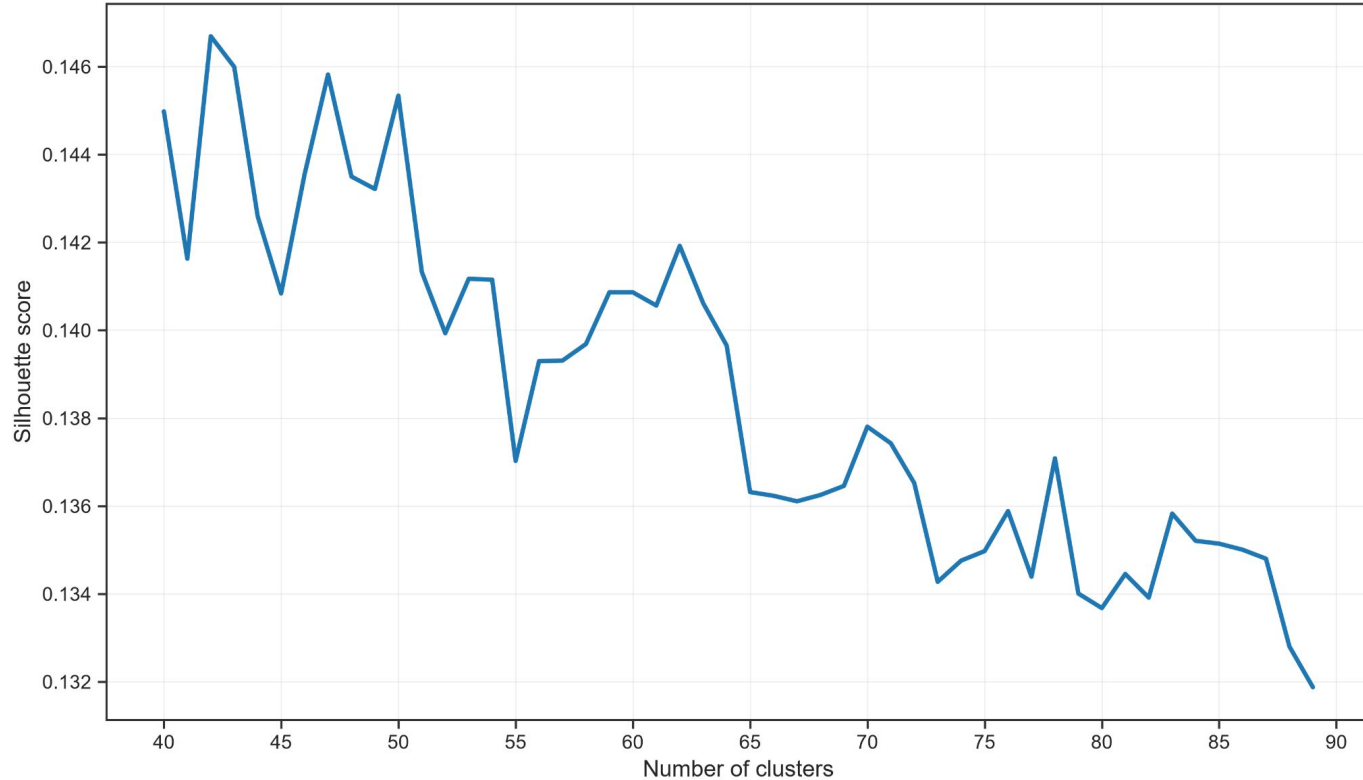
No clear elbow 'k' value



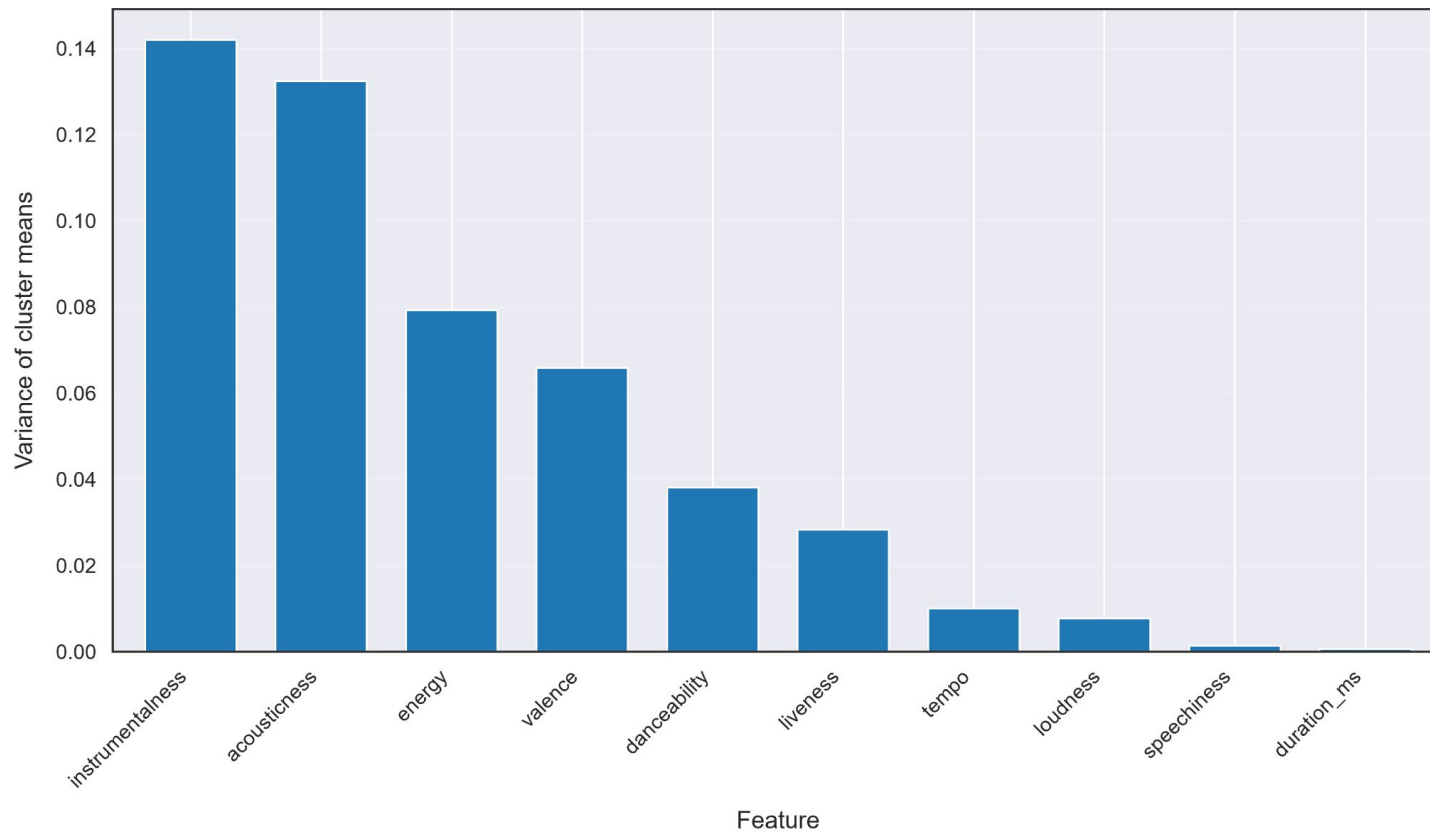
Silhouette Scores



Silhouette scores also do not inform much



Feature Variances



Russell's circumplex model

